ERAM RAHMAN

US Citizen | San Diego, CA (UC San Diego) | Los Angeles, CA | Open to Relocation | +1 747-955-6221 |
eramrahman674@gmail.com | https://eramrahman2831.github.io/portfolio/

EDUCATION

Expected Graduation: 2027

Bachelor of Science in Computer Engineering — University of California San Diego
Major: Computer Engineering
Relevant Coursework: Data structures and Algorithms, Software Engineering, Digital Systems, Computer Architecture, Operating Systems

SKILLS

- **Programming Languages:** Python, C++, C, Assembly (ARM), HTML, CSS, Java, JavaScript, MATLAB, TypeScript
- **Web and Application Development:** AWS, React, Next.js, JavaFX, SQL, SQLite, Git/GitHub, REST APIs, Node.js, Anthropic API, Agile Development, Flask and UI-based automated testing
- **IT and Hardware:** PC Assembly, Hardware Troubleshooting, System Diagnostics

EXPERIENCE

GiosTech Software Engineering Internship | *Python, Java, Anthropic API* **March 2026 - June 2026**

- Developed a full-stack web application using Three.js and a Python/Flask backend to render, manipulate, and interact with 3D human muscle models.
- Created an interactive AI chatbot using the Anthropic API to collect user-reported pain information, identify likely affected muscle groups, and recommend relevant at-home exercises.

Undergraduate SLAM Researcher | *Python, HTML* **January 2026 - Present**

- Developed a robotic arm simulation to test motion planning and pathfinding algorithms.
- Led development efforts for a research workshop, coordinating tasks and guiding team members to complete a technical paper and presentation on robot perception and planning, increasing efficiency by 30%.

PC Technician and Owner **February 2019 - Present**

- Provided technical consulting and support to 100+ clients, diagnosing hardware and software issues with a high success rate.
- Reduced client downtime by 30–50% through efficient troubleshooting and hardware optimization.
- Provided technical support and hardware recommendations.

PROJECTS

PC Recommendation Website | *React, Next.js*

- Developed a full-stack web application using Next.js (React), achieving perfect Google Lighthouse scores (100/100/100)
- Designed a recommendation algorithm that evaluates user survey responses to identify optimal PC and laptop options based on budget, workload, and hardware performance requirements.
- Integrated external APIs (Canopy API) to fetch product data and implemented server-side filtering on around 20 products per request based on user survey inputs.

Voice Controlled Navigation | *React, CSS, HTML*

- Built a web-based navigation application using Mapbox API with real-time GPS tracking and routing, achieving 150–170 ms API response times and a Google lighthouse performance score of 100.
- Implemented voice command functionality for hands-free navigation, reducing manual interaction by 25–35% and improving user safety.
- Developed dynamic route recalculation and multi-stop navigation, thus improving routing efficiency.

Student Clock Application | *SQLite, Agile Development, Java, CSS, HTML, JavaFX*

- Developed a Java-based productivity application using JavaFX and SQLite for task and time management.
- Implemented priority-based scheduling algorithm to optimize study time allocation by about 50%.
- Implemented automated UI tests with Jest and React Testing Library to validate including clock-in/clock-out actions, schedule rendering, and form input validation.